



COMPONENT LIFECYCLE IN ANGULAR



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✓ What is the Component Lifecycle?

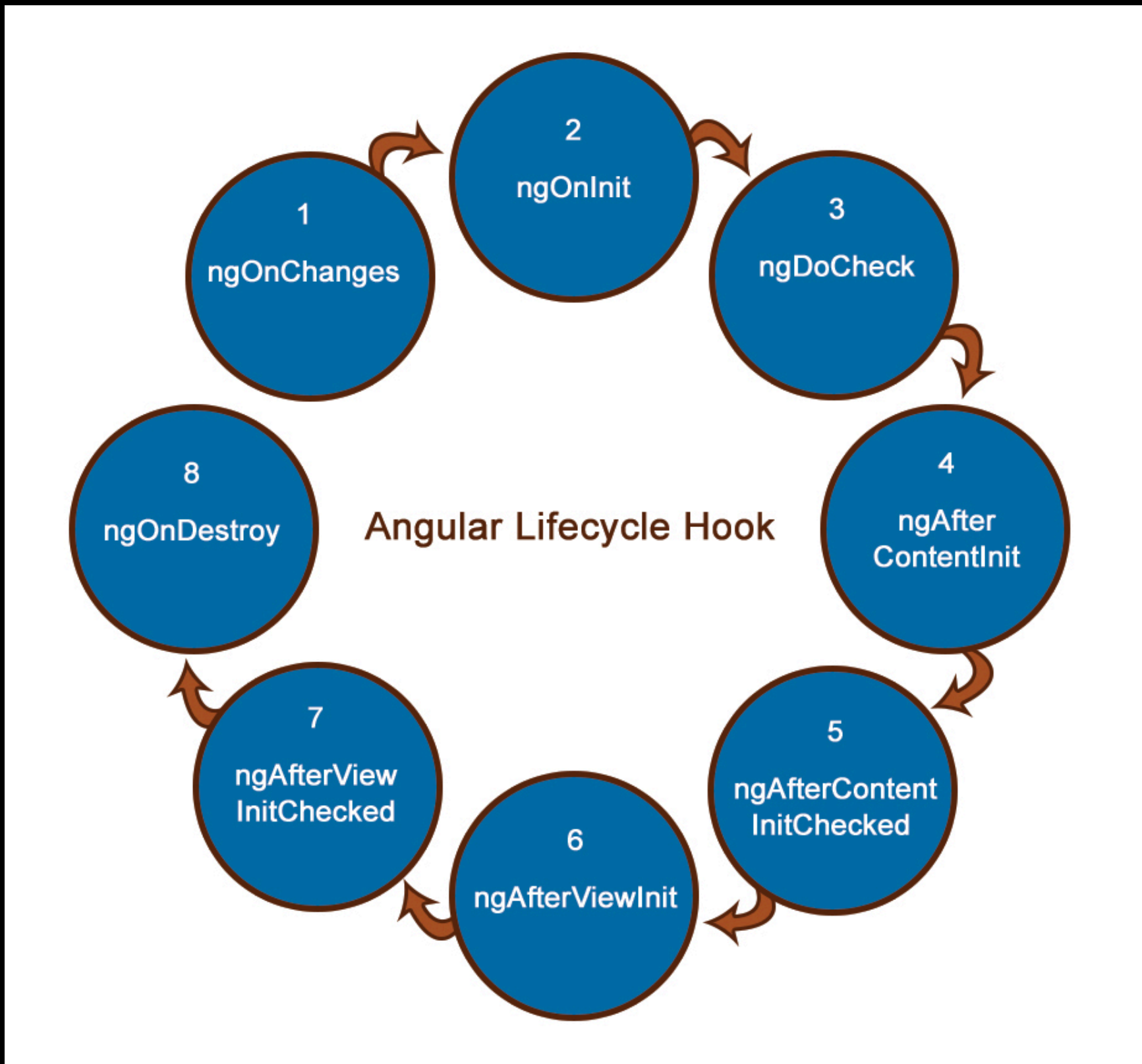
A **component instance** has a lifecycle that starts when Angular instantiates the **component class** and **renders the component view** along with its child views.

📌 Helps with:

- DOM initialization
- Data fetching
- Cleanup
- Debugging component behavior

📌 Angular components don't just appear they follow a lifecycle managed by Angular.

✓ Lifecycle Hook Overview



Here are the major lifecycle hooks in the order they're called:

- 1. ngOnChanges()**
- 2. ngOnInit()**
- 3. ngDoCheck()**
- 4. ngAfterContentInit()**
- 5. ngAfterContentChecked()**
- 6. ngAfterViewInit()**
- 7. ngAfterViewChecked()**
- 8. ngOnDestroy()**

Each serves a specific purpose in a component's lifespan.

✓ ngOnChanges()

When? → On every change to input-bound properties.



app.component.ts

```
ngOnChanges(changes: SimpleChanges) {  
  console.log('Input changed:', changes);  
}
```

- ◆ Called before ngOnInit
- ◆ Useful for reacting to @Input() changes
- ◆ Gets a SimpleChanges object

✓ **ngOnInit()**

When? → Once, after the first ngOnChanges().

```
● ● ● app.component.ts  
  
ngOnInit() {  
  this.fetchData();  
}
```

Best for:

- Initial data loading
- Setting up component state
- Subscribing to observables

✓ ngDoCheck()

When? → Custom change detection logic.

```
● ● ● app.component.ts  
  
ngDoCheck( ) {  
    // Custom change detection logic
```

- **Called during every change detection cycle**
- **Use cautiously — it runs frequently**

View & Content Hooks

ngAfterContentInit()

- ◆ Called after Angular projects external content into component.

ngAfterViewInit()

- ◆ Called after component's view (and child views) are initialized.

Great for:

- DOM querying
- ViewChild/ContentChild access
- Running animations

✓ **ngOnDestroy()**

When? → Just before Angular destroys the component.

```
app.component.ts

ngOnDestroy() {
  this.subscription.unsubscribe();
}
```

Use for:

- **Clearing timers**
- **Unsubscribing from observables**
- **Cleaning memory leaks**
- **Detaching event listeners**

✓ Real-World Usage Example

```
app.component.ts

export class ProductComponent implements OnInit, OnDestroy {
  subscription: Subscription;

  ngOnInit() {
    this.subscription = this.dataService.getProducts().subscribe();
  }

  ngOnDestroy() {
    this.subscription.unsubscribe();
  }
}
```

🎯 This prevents memory leaks and improves performance.

✓ **Real-World Usage Example**

- **Always unsubscribe in ngOnDestroy()**
- **Use ngOnInit() for initialization logic**
- **Avoid heavy logic in ngDoCheck()**
- **Don't manipulate DOM before
ngAfterViewInit()**
- **Understand the sequence to avoid
bugs**



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